5

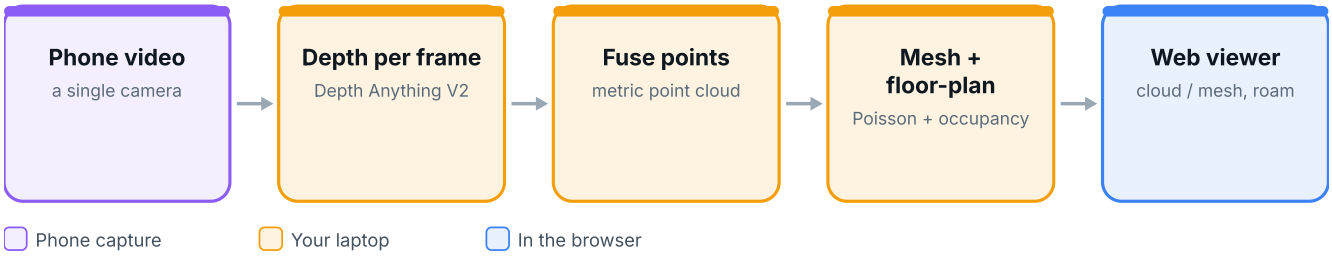
OPTION 5 · RECONSTRUCTION

monomap — our own pipeline

Mono-depth AI → point cloud + mesh + floor-plan, fully offline.

Our build · fallback + R2

In one line: our own pipeline — estimate **depth** from each video frame with an AI model, fuse into a 3D point cloud + mesh + **floor-plan**, all **offline on your laptop**. Not photorealistic, but it's ours end-to-end and it already runs.



100% offline on your laptop. Uniquely also outputs an occupancy grid + floor-plan — the seed of Round-2 navigation.

How it works, step by step

1. Split the **phone video** into frames.
2. **Depth Anything V2** predicts a depth map for every frame (metric indoor model → real metres).
3. Back-project + **fuse** the depths into one **metric point cloud** (~3.9 M points) with a camera path.
4. Build a **Poisson mesh**, detect planes, and rasterise a **floor-plan + 3D occupancy grid**.
5. View the colored cloud / mesh in the browser (and, for Round 2, the floor-plan for pathfinding).

Trade-offs

✓ Strengths

- Fully **offline** — no cloud, no account, private
- Our own code — strongest 'pipeline soundness' story
- Outputs a **floor-plan + occupancy map** (nav-ready)
- Already working: cloud + mesh + floormap produced

✗ Weaknesses

- **Not photorealistic** — mono-depth is noisy
- Poisson mesh looks webby / imperfect
- Needs the 4 GB GPU (Smart App Control sometimes blocks it)
- Weaker on the 'visual fidelity' score than splats

Tools: Depth Anything V2, Open3D | Input: phone video | Output: cloud + mesh + floor-plan | Your compute: laptop (offline) | Cost: free

WHAT WE ACTUALLY DID

Built and working. Produced a 3.9 M-point metric cloud, a 400 K-triangle mesh, detected planes, and a ROS-style floor-plan + 3D occupancy grid from `clip1.mp4`. We also built an offline browser viewer for it.

VERDICT FOR THE COMPETITION

Our safety net for Round 1 (ship it if no good splat lands) and the **navigation engine for Round 2** — the floor-plan + occupancy grid are exactly what obstacle-aware routing needs.